SWEET NOTE

MUSIC ACADEMY

Yasmin Shawamreh

DBA210 Final Project

ABSTRACT

This Project Report introduces Sweet Note Music Academy, a music school that offers one on one private lessons and performance reviews to help students focus on mastering their performance skills. The Sweet Note Music Academy database system (sn) will be explained thoroughly to clarify management of students, instructors, instruments, and other music school details through the MySQL platform. This database system will allow Sweet Note Music Academy to retrieve any data necessary to make sure the academy runs smoothly.

INTRODUCTION

Sweet Note Music Academy is a music school that offers private lessons to all ages 5 years old and above. Sweet Note offers lessons in piano, guitar, drums, violin, voice, and so much more! They have incredible instructors, most are multi-instrumentalists, who are dedicated to ensuring that students receive the time and care it takes to learn an instrument fluently.

Financiers and students will be able to choose from multiple Sweet Note music lesson packages to bundle and save money on lessons. Each quarter, our instructors will ask that students pick a venue of their choice from our performance venues list to perform and send the recording of their performance to their instructors to be graded. This will document each students' growth and can be used in their portfolios to further their careers in music.

BUSINESS REQUIREMENTS

Sweet Note Music Academy has requested the following business requirements:

Students must be 5 years old or older.

By adding and altering the students table and giving the student's age a constraint, students entered are only allowed to be 5 years old or older.

Students are allowed to learn only one instrument at a time to ensure focus and precision.

The Students and Instruments tables are connected through a one to one relationship ensuring that students are only able to sign up for one instrument at a time.

Financiers must purchase packages as single lessons are not offered.

Sweet Note packs, which include the pricing and amount of lessons per pack, are available for the financier to purchase. These packs are connected to the invoice packages table which allows for invoices to have as many Sweet Note packs as requested.

Each instructor must deliver a performance review to each student quarterly.

The Instructors table has a one to many relationship with the performance reviews table allowing each student to receive as many reviews as needed.

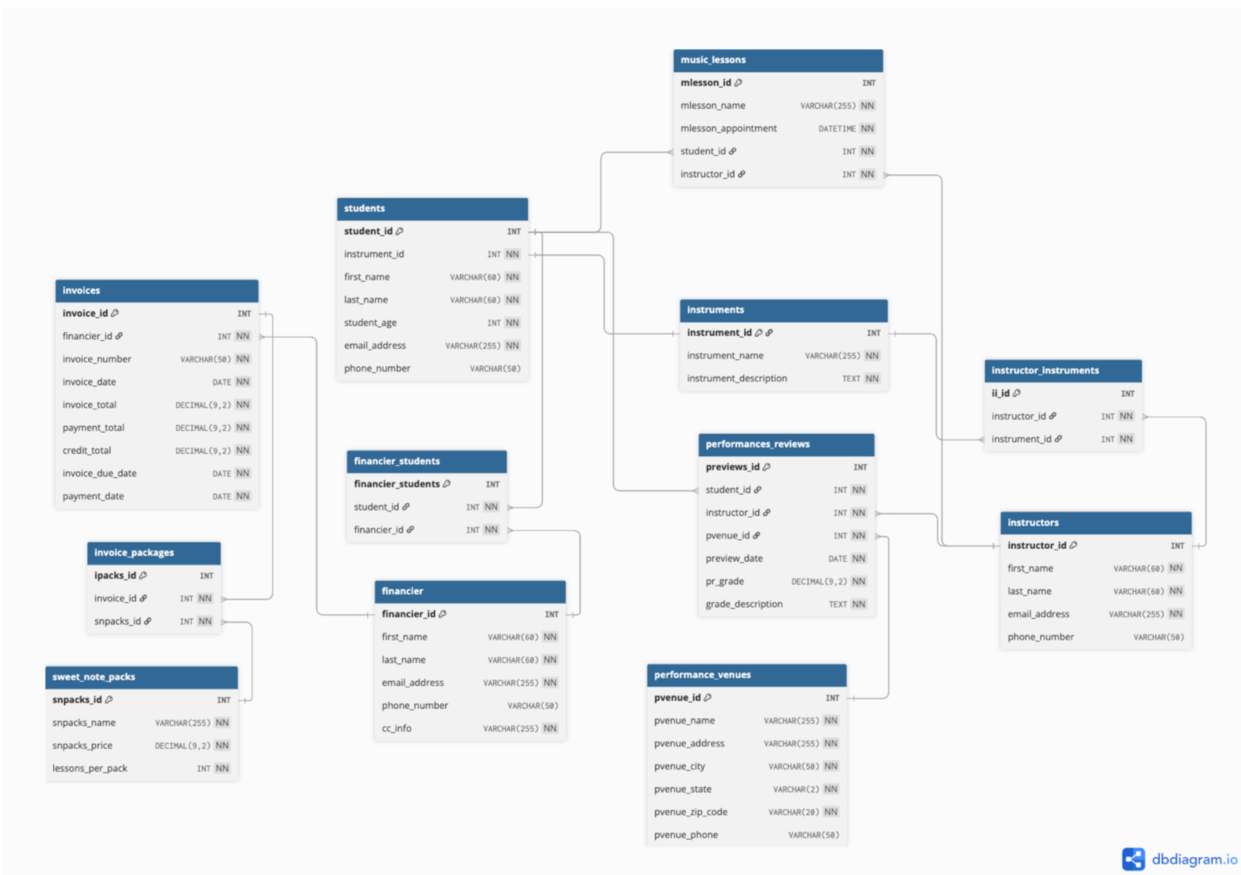
Each student must select performance space from provided venues list to record and send performance to instructor for grading and reviews.

Both the Students and Instructors tables are connected to the Performance Reviews table. The Performance Venues table is available for students to select their venue of choice.

ERD

[Sweet Note Music Academy - Entity Relationship Diagram or ERD]:

https://dbdiagram.io/d/Sweet-Note-Music-Academy-687e7aa5f413ba3508ef4702



Sweet Note Music Academy ERD contains the following tables:

“instruments”, “students”, “instructors”, “financier”, “financier students”, “instructor instruments”, “music lessons”, “sweet note packs”, “invoices”, “invoice packages”, “performance venues”, “performance reviews”

TABLE DEFINITION

INSTRUMENTS

Data required for the instruments table: instrument ID, instrument name, and instrument description.

STUDENTS

Data required for the students table: student ID, instrument ID, first name, last name, student's age, email address, and phone number.

INSTRUCTORS

Data required for the instructors table: instructor ID, first name, last name, email address, and phone number.

FINANCIER

Data required for the financier table: financier ID, first name, last name, email address, phone number and credit card information.

FINANCIER STUDENTS

Data required for the financier students table: financier students ID, student ID and financier ID.

INSTRUCTOR INSTRUMENTS

Data required for the instructor instruments table: instructor instruments (ii) ID, instructor ID, and instrument ID.

MUSIC LESSONS

Data required for the music lessons table: music lesson ID, music lesson name, music lesson appointment date and time, student ID, and instructor ID.

SWEET NOTE PACKS

Data required for the sweet note packs table: sweet note packs ID, sweet note packs name, sweet note packs price, and lessons per pack.

INVOICES

Data required for the invoices table: invoice ID, financier ID, invoice number, invoice date, invoice total, payment total, credit total, invoice due date, and payment date.

INVOICE PACKAGES

Data required for the invoice packages table: invoice packages ID, invoice ID, and sweet note packs ID.

PERFORMANCE VENUES

Data required for the performance venues table: performance venue ID, performance venue name, performance venue address, performance venue city, performance venue state, performance venue zip code and performance venue phone number.

PERFORMANCE REVIEWS

Data required for the performance reviews table: performance reviews ID, student ID, instructor ID, performance venue ID, performance review date, performance review grade, and the grade description.

DATA NORMALIZATION

Data normalization is the process of organizing a database in a way that minimizes data redundancy and enhances data retrieval efficiency. In First Normal Form (1NF), every cell would hold just one value so that there are no lists or sets inside a column. In Second Normal Form (2NF), a table must already by in 1NF and all non-key columns must be fully dependent on the entire primary key.

In this project, the database has been structured to satisfy the requirements of Third Normal Form (3NF)**,** which is widely recognized as the standard for a fully normalized database. To meet 3NF, each table includes a primary key, typically named after the table itself (for example: "students" table – primary key: "student_id"), and all non-key attributes are dependent solely on that primary key.

Sweet Note Music Academy requested the following ERD (Database Schema) requirements:

Students can have multiple private music lessons.

The Students table has a one to many relationship with the Music Lessons table.

Instructors can teach multiple private music lessons.

The Instructors table has a one to many relationship with the Music Lessons table.

Instructors can teach multiple instruments, but students can only learn one instrument at a time.

The students table has a one to one relationship with the Instruments table. The Instructors table and the Instruments table have a many to many relationship and shares the join table Instructor Instruments to make sure instructors can teach multiple instruments.

The financier can be a parent, guardian, or student, can have multiple invoices and can finance more than one student at a time.

The Financier and Students table share a many to many relationship joined by the Financier Students table. The financier table has a one to many relationship with the Invoices table.

Each invoice can have multiple sweet note packages assigned.

The Invoices and Sweet Note Packs tables share a many to many relationship which is joined by the Invoice Packages table.

Each student will have multiple performance reviews.

The Students table has a one to many relationship with the Performance Reviews table.

Each venue will have multiple performances for reviews.

The Performance Venues table has a one to many relationship with the Performance Reviews table.

Each performance review is associated with one instructor.

The Instructors table has a one to many relationship with the Performance Reviews Table.

CONCLUSION

With the implementation of the Sweet Note database using MySQL, Sweet Note Music Academy can efficiently organize and manage its operations through a well-structured and relational system. This database allows for the management of students, instructors, instruments, performances, and financial transactions.

The aim for this final project was to provide clarity and structure behind the Sweet Note database. By applying First, Second and Third Normal Forms, data redundancy has been minimized, and data integrity is maintained across all related tables.

This final project shows how a well-structured database can improve operational efficiency and provide a solid foundation for managing a music academy.